

Element H

The handle for the cart will work if it meets the criteria and requirements that we set. Our Criteria includes, fit, ease of use, removability, strength, and impact on the cart.

Fit: The handle must make the cart comfortable to use for Mr Kiser. This is the point of building the handle so if it doesn't work for Mr. Kiser the handle is useless.

Ease of Use: The handle must be simple and intuitive to use. If the handle has a complicated method of adjusting it or removing it will be less desirable for the client. The removability is a large part of this design therefore it needs to be a simple method to connect and remove the handle.

Removability: The handle must be easily removable from the cart. From the research we conducted and the interview we had, the handle will not be used if it is difficult to remove as the handle is not always needed on the cart.

Strength: The handle must be able to withstand the normal use of the cart and the maximum load of the cart. If the cart can't be fully loaded the usefulness of the handle is decreased greatly. The handle also needs to last normal use of the cart for years with minimal to no replacement parts. If the handle constantly needs fixed and is broken it is of no use to any clients.

Impact on Cart: The handle must not require permanent modification to the cart in any way nor must it reduce the area of the cart which is usable for materials. Based on our interview the handle is a useless solution if it requires permanent changes to the cart. The solution must also not reduce the area of the shelves to place objects on as the full area of the cart is needed to move certain items.

Ideal Results

Meeting all ideal requirements with the product through testing would indicate that the solution met if not surpassed all criteria and requirements of the project.

Fit: The handle is a comfortable fit for Mr. Kiser and 90% of all people remove any back pain when pushing the cart fully loaded.

Ease of use: The handle is intuitive for all users in the way that it is used as a handle as well as the way that the handle removes requiring zero instructions.

Removability: The handle must be able to attach and remove from the cart in under 30 seconds without the user needing to bend down.

Strength: The handle is able to push 500lbs up a slope with no destructive effect on the handle allowing it to be used in this way constantly.

Impact on Cart: The cart must not have any marks to tell the handle had been there after the handle is removed.

Minimum Requirements

Meeting the minimum requirements would indicate that the product works for the intended purpose and is effective, however not as well as intended.

Fit: To meet the minimum requirements the handle must allow Mr. Kiser to push the cart for longer periods of time with reduced back pain.

Ease of Use: To meet the minimum requirements the handle is intuitive in its use and requires minimal directions for the user to remove it from the cart.

Removability: To meet the minimum requirements the handle can be attached or removed in less than one minute with the user having to bend down minimally

Strength: To meet the minimum requirements the handle can withstand typical use with light loads constantly and occasional heavy loads (200-500 lbs).

Impact on the Cart: To meet the minimum requirements the function of the cart, with the handle removed, is unaffected with minimal visual effects upon the cart.